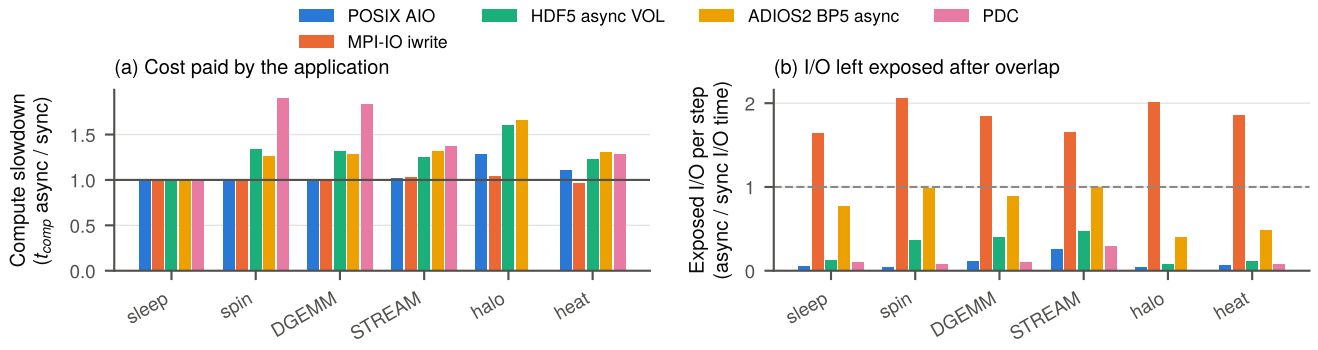
POSIX AIO
MPI-IO iwrite
HDF5 async VOL
ADIOS2 BP5 async
PDC
(a) Cost paid by the application
Compute slowdown
(t_{comp} async / sync)
0.0
0.5
1.0
1.5
sleep
spin
DGEMM
STREAM
halo
heat
(b) I/O left exposed after overlap
Exposed I/O per step
(async / sync I/O time)
0
1
2
sleep
spin
DGEMM
STREAM
halo
heat